

Unit 7. Law of Large Numbers and Central Limit Theorem

Random samples, Chebyshev, LLN, CLT

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 15 and 17, CC BY-NC-SA 4.0.

Roadmap

- 1 Random samples and the sample mean
- 2 Chebyshev's inequality
- 3 Law of Large Numbers
- 4 Convergence in distribution
- 5 Central Limit Theorem
- 6 Bridge to estimation and inference
- 7 Wrap-up

From distributions to data

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Mean and variance: $\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \sigma^2/n.$

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$\text{Var}(\bar{X}_n) \rightarrow 0$ is the seed of the LLN.

Chebyshev's inequality

Theorem

For any random variable X with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) < \infty$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2}.$$

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Reading. Variance $\rightarrow$ deviation probability. Makes the slogan “variance measures dispersion” literal. **Applied to the sample mean:**

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n \varepsilon^2} \xrightarrow{n \rightarrow \infty} 0.$$

Convergence in probability

Definition

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“ Y_n is, with high probability, arbitrarily close to c for large n .”

Weak Law of Large Numbers

Theorem

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One-line proof. Chebyshev: $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0.$

Weak Law of Large Numbers

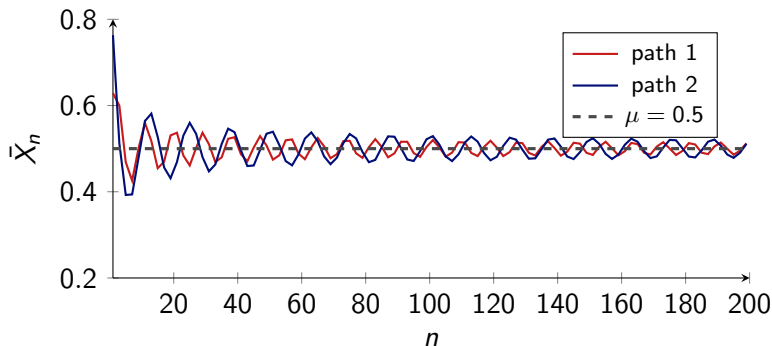
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One-line proof. Chebyshev: $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$. **Slogan.** Sample averages estimate population means. This is the formal justification for using $\bar{X}_n$ as an estimator of μ — the seed of consistency in Unit 9.

LLN in pictures



Two simulated paths of the running mean of i.i.d. Bernoulli(0.5) draws. Both **collapse** onto $\mu = 0.5$ as n grows.

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Lesson

Even with mostly uninformed voters, **independent** noise averages out — and large electorates choose well.

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Lesson

Common shocks invalidate the LLN. This is why econometricians worry about cross-sectional dependence and cluster their standard errors.

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$Y_n \xrightarrow{p} c \Rightarrow Y_n \xrightarrow{d} c$, but not the converse.

Central Limit Theorem

Theorem

$X_1, \dots, X_n$ i.i.d. with $\mathbb{E}[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2 \in (0, \infty)$. Then

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

Equivalently, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$.

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Why normal?

- Sums of independent normals are normal — the family is **stable** under independent addition.
- Among finite-variance families, normal is the unique stable one.
- Hence the rescaled sample mean is attracted to it.

LLN and CLT live together

Same object, two views.

$$\bar{X}_n = \mu + \frac{\sigma}{\sqrt{n}} Z_n + o_p(1/\sqrt{n}), \quad Z_n \xrightarrow{d} \mathcal{N}(0, 1).$$

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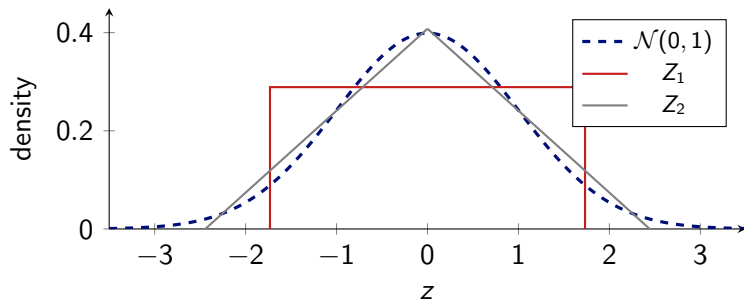
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- First term $\Rightarrow$ LLN: $\bar{X}_n \rightarrow \mu$.
- Second term $\Rightarrow$ CLT: residual is $\sqrt{n}$ -normal.

$\sqrt{n}$ is the unique rescaling that keeps the limit non-degenerate and bounded.

CLT in action: averaging uniforms



$$X_i \stackrel{\text{iid}}{\sim} U[0, 1], \mu = \frac{1}{2}, \sigma^2 = \frac{1}{12}.$$

For $n \geq 30$, the standardised mean of i.i.d. uniforms is **visually indistinguishable** from the standard normal.

Practical guidance and caveats

- **Rule of thumb:** $n \geq 30$ for “well-behaved” parents.
- Skewed/fat-tailed parents (e.g. χ_1^2 , lognormal) need larger n .
- **No finite variance, no CLT.** Cauchy parents: $\bar{X}_n$ is itself Cauchy for every n — LLN *and* CLT both fail.

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Why the CLT is so useful

- ① We don't need to know f_X exactly — only μ and σ^2 .
- ② Normal probabilities are tabulated.
- ③ All standard test statistics in econometrics rest on a CLT.

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- ① **Consistency:** $\hat{\theta}_n \xrightarrow{p} \theta_0$ (Unit 9 — LLN feeds this).
- ② **Inference:** confidence intervals built from the normal table (Unit 10 — CLT feeds this).
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The OLS estimator $\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{Y}$ is asymptotically MVN with covariance $\sigma^2 (X^\top X)^{-1}$ — a multivariate CLT statement.

Take-aways

- ① Random samples are i.i.d. draws from f_X . The sample mean has mean μ and variance σ^2/n .
- ② **Chebyshev** bounds deviation probabilities by variance.
- ③ **LLN**: $\bar{X}_n \xrightarrow{p} \mu$. Independence of the noise is what makes it work.
- ④ **CLT**: $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$. The unique stable, finite-variance limit.
- ⑤ LLN + CLT are the two pillars on which estimation and inference rest.

Next unit

Point estimation, MLE, Method of Moments, and OLS.

Thank you

Questions?

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